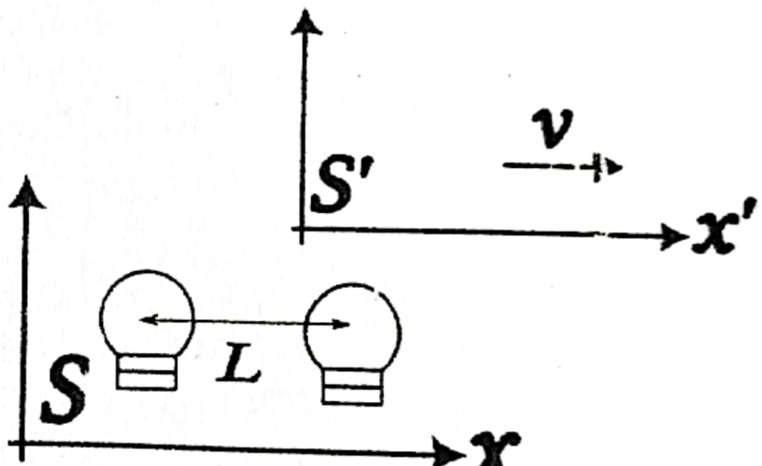
MID SEMESTER TEST - 2

B.TECH. PH10009: APPLIED PHYSICS

Time : 1 hour | Date: November 04, 2025 | Max Marks : 20

Given: $\hbar = 1.054 \times 10^{-34}$Js and $m_e = 9.11 \times 10^{-31}$kg.

S.No.	Question	Mark	CO	BL
1	As shown in figure, S and S' are two inertial frame of references. Two lamps, separated by a distance L in S are switched 'ON' simultaneously. In S' an observer measures the distance between the lamps to be L' and observes the lamps go 'ON' with the time difference τ. Express L in terms of L' and τ. Also find an expression for the relative velocity v. 	4	3	2

[P T O]

2	(i) Obtain Lorentz transformation equations for space (x, y, z and x', y', z') and time (t and t'). (ii) What do you understand from the following terms (a) Simultaneity (b) Frame of reference (c) inertial frame of reference and (d) time dilation.	3+3	3	2
3	Discuss an experiment (studied in Quantum Mechanics), assuming light as a particle. Derive an expression that proves and validate the experimental results.	1½+4	1	1
4	Find the following (i) Wavelength of a person of mass 50kg moving with a velocity of 10cm/s, (ii) Wavelength (in Å) of an electron accelerated under a potential of 100V. (iii) If, the uncertainty in the position of an electron is 7.9×10^{-12}m, find the uncertainty in the velocity of the electron.	3×1½	2	5